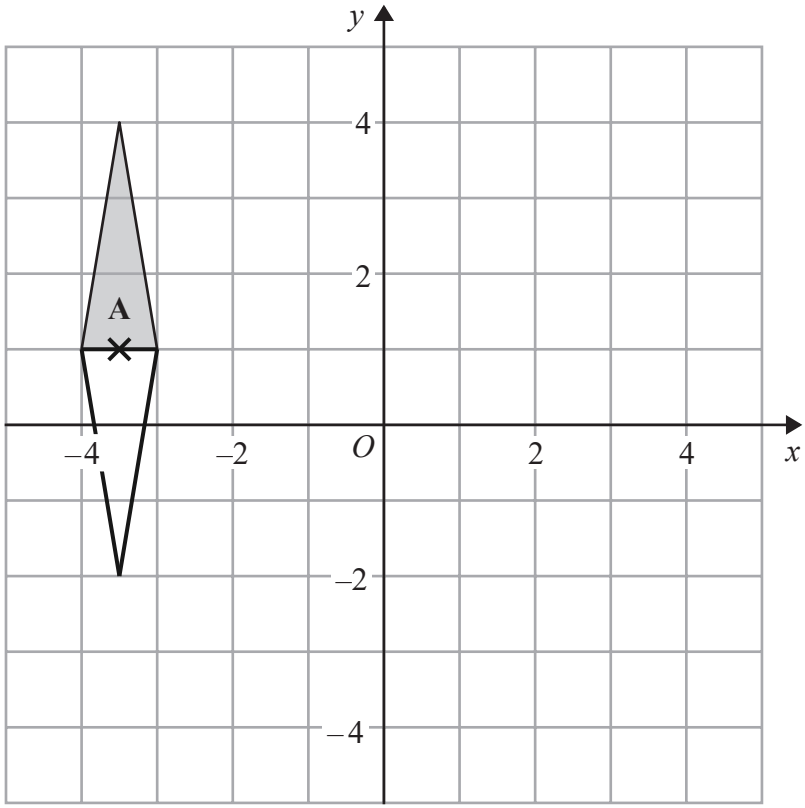
20



Rotate:

$x' = -4 - x$

$y' = -y$

Then translate:

$x'' = -7 - x$

$y'' = 2 - y$

1 mark
Apply both transformations.

Triangle **A** is transformed by the combined transformation of a rotation of 180° about the point (−2, 0) followed by a translation with vector $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

One point on triangle **A** is invariant under the combined transformation.

Find the coordinates of this point.

Invariant: $x = -7 - x$ and $y = 2 - y$

$2x = -7$, $2y = 2$, so $x = -3.5$, $y = 1$.

(−3.5 , 1)

1 mark
Correct invariant point.

(Total for Question 20 is 2 marks)